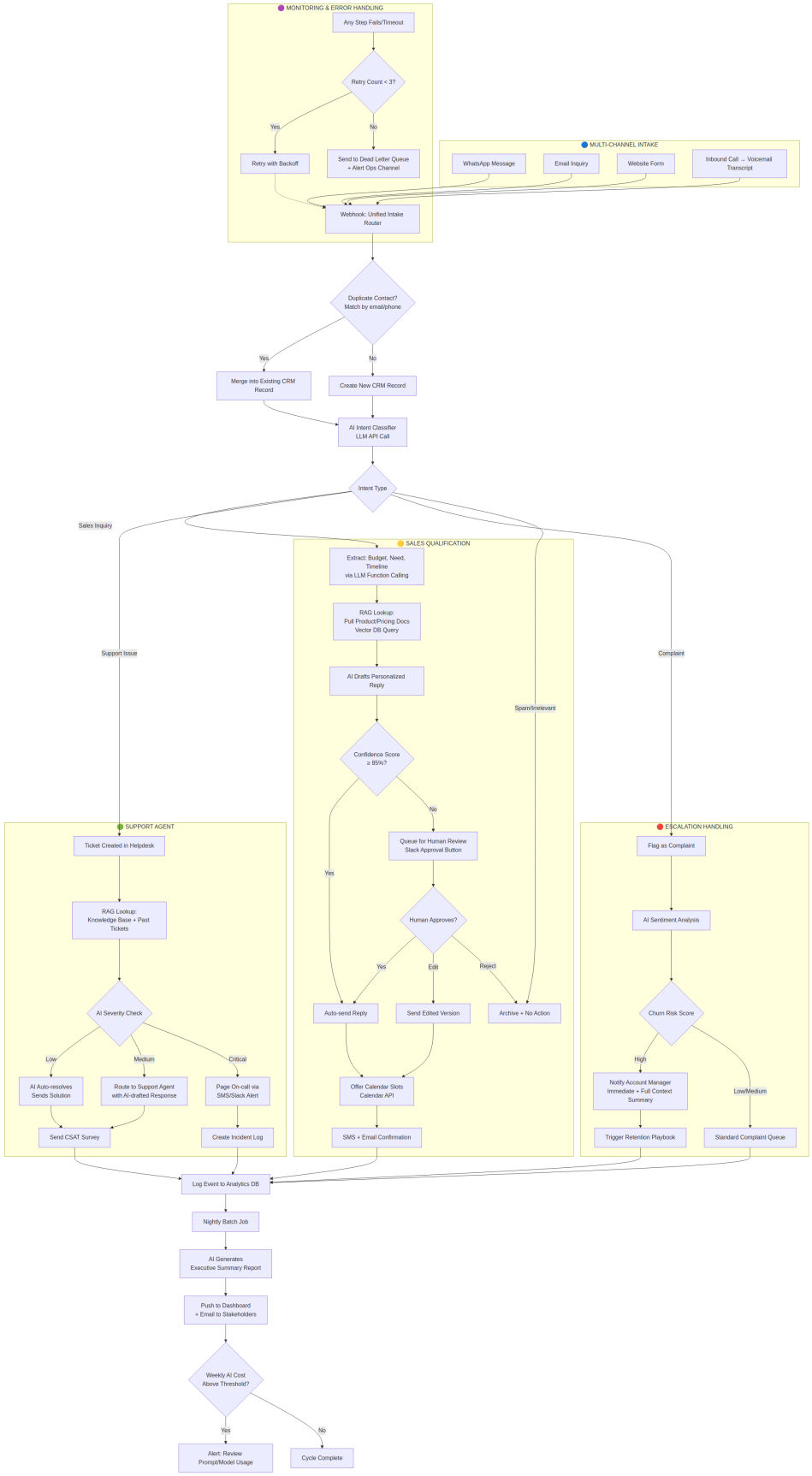
Advanced AI Automation Workflow

Multi-Channel Intake → AI Routing → Sales / Support / Escalation → Monitoring & Reporting


🟣 MONITORING & ERROR HANDLING
Any Step Fails/Timeout
Retry Count < 3?
Yes
No
Retry with Backoff
Send to Dead Letter Queue + Alert Ops Channel
🔵 MULTI-CHANNEL INTAKE
WhatsApp Message
Email Inquiry
Website Form
Inbound Call → Voicemail Transcript
Webhook: Unified Intake Router
Duplicate Contact? Match by email/phone
Yes
No
Merge into Existing CRM Record
Create New CRM Record
AI Intent Classifier LLM API Call
Intent Type
Sales Inquiry
Support Issue
Spam/Irrelevant
Complaint
🟡 SALES QUALIFICATION
Extract: Budget, Need, Timeline via LLM Function Calling
RAG Lookup: Pull Product/Pricing Docs Vector DB Query
AI Drafts Personalized Reply
Confidence Score ≥ 85%?
Yes
No
Queue for Human Review Slack Approval Button
Human Approves?
Yes
Edit
Reject
Auto-send Reply
Send Edited Version
Archive + No Action
Offer Calendar Slots Calendar API
SMS + Email Confirmation
🟢 SUPPORT AGENT
Ticket Created in Helpdesk
RAG Lookup: Knowledge Base + Past Tickets
AI Severity Check
Low
Medium
Critical
AI Auto-resolves Sends Solution
Route to Support Agent with AI-drafted Response
Page On-call via SMS/Slack Alert
Send CSAT Survey
Create Incident Log
🔴 ESCALATION HANDLING
Flag as Complaint
AI Sentiment Analysis
Churn Risk Score
High
Low/Medium
Notify Account Manager Immediate + Full Context Summary
Trigger Retention Playbook
Standard Complaint Queue
Log Event to Analytics DB
Nightly Batch Job
AI Generates Executive Summary Report
Push to Dashboard + Email to Stakeholders
Weekly AI Cost Above Threshold?
Yes
No
Alert: Review Prompt/Model Usage
Cycle Complete

Exact Tools Used

Component	Tool
Orchestration engine	n8n (self-hosted, handles Code nodes + complex branching)
WhatsApp intake	Twilio WhatsApp API / WhatsApp Business API
Email intake	Gmail / Outlook API or Postmark webhook
Website form	Native Webhook node (n8n)
Voicemail transcript	Twilio Voice + OpenAI Whisper API (speech-to-text)
CRM dedupe / create	HubSpot or Airtable API (HTTP Request node)
Intent classifier	OpenAI / Claude API - function calling, structured JSON output
RAG lookup (docs / KB)	Pinecone or Chroma (vector DB) + OpenAI Embeddings API
Confidence-gated auto-send	Python Code node in n8n - threshold logic
Human approval step	Slack API - Interactive Buttons (n8n native Slack node)
Calendar booking	Cal.com API or Google Calendar API
SMS / email confirmations	Twilio (SMS) + Postmark / SendGrid (email)
Helpdesk ticketing	Zendesk or Freshdesk API
Severity / sentiment analysis	Claude / OpenAI API (structured output)
On-call paging	Twilio SMS or Slack webhook to on-call channel
Retry / error handling	n8n built-in Error Trigger node + Code node retry logic
Dead letter queue	Airtable / Google Sheet or Postgres table
Analytics / event log	Airtable, Google Sheets, or Postgres (HTTP / DB node)
Nightly summary report	n8n Schedule Trigger (cron) + Claude/OpenAI API + SendGrid
Cost monitoring	Custom Python script tracking token usage, Code node comparison

Note: n8n / Make / Zapier serve as the orchestration layer. SMS, calendar, CRM, and vector database services are external systems connected into the orchestration tool via native integrations, webhooks, or HTTP API calls.